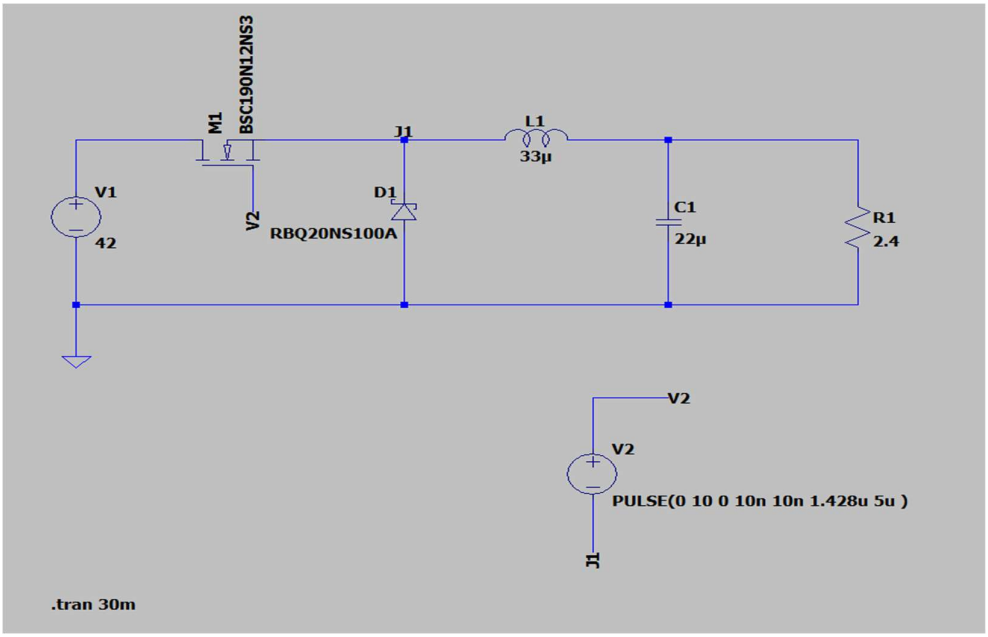
Buck converter design document for ideal case (Version 1)

1. **Objective:** Design and analyze a 60W buck converter to step down input voltage from 42 V to 12 V using LT Spice.

2. **Specifications :**
   - Input voltage(Vin): 42 V
   - Output voltage(Vout): 12 V
   - Output Current(Iout): 5 A
   - Output power(Pout): 60 W
   - Switching frequency(Fsw): 200KHz
   - Load resistance(Rl): 2.4Ω

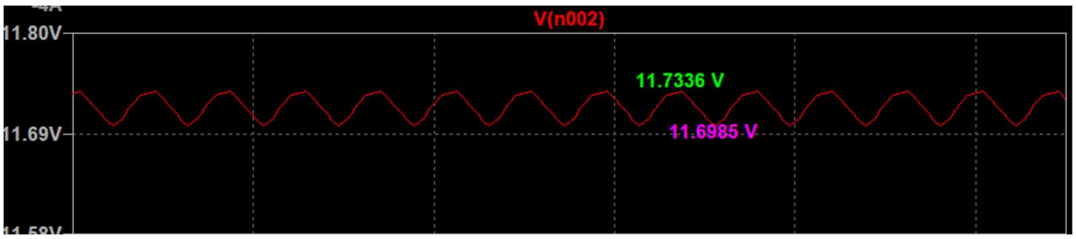
3. **Design calculation (Ideal):**
   - Duty Cycle(D1): Vout/Vin: 12/42 = 0.2857
   - Inductor Selected: 33µH
   - Estimate Inductor ripple: 1.4A – 1.6A (~30% of Imax)
   - Capacitor selected: 22µF
   - MOSFET selected: BSC190N12NS3(Infineon)
   - Diode selected: Schottky (RBQ20NS100A – Rohm)
   - CCM Boundary check: L critical = (1-D) * RL / (2fs) = 4.28µH. Here 33µH >> Lcritical Converter operates CCM
   - **MOSFET and diode modelled using real manufacturer SPICE models. Inductor and capacitor modelled as ideal elements – no parasitics involved in this version 1.**

4. **Circuit Visual:**

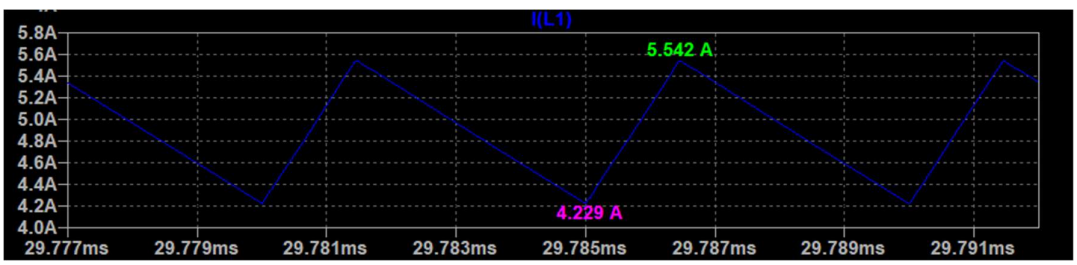


5. Simulation Results:

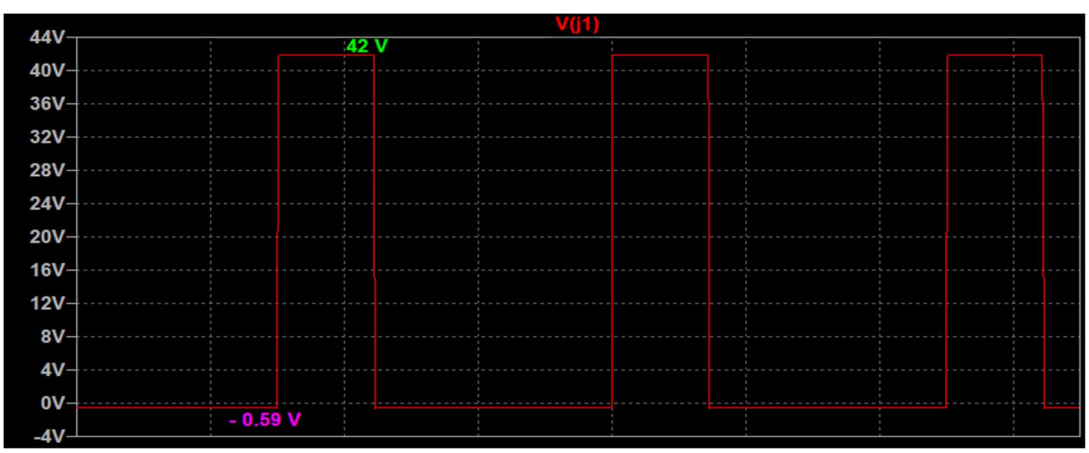
- **Actual output (Vout):** Voltage varies from 11.698V to 11.734V with an average voltage 11.716V (The deviation from ideal 12V is due to forward voltage drop of Schottky diode (~0.75 to 0.8V) and associated conduction loss resulting in approximately 2.5-3W power dissipation. While the observed ~35mv peak to peak ripple is due to inductor ripple current charging and discharging the output capacitor).



- **Inductor current (IL):** Current varies from 4.229A to 5.542A with an average current 4.885A which closely matches with the output current 5A. The ripple current is approximately 1.31A (~26%), confirming that the converter operates in continuous conduction mode (CCM) as expected.



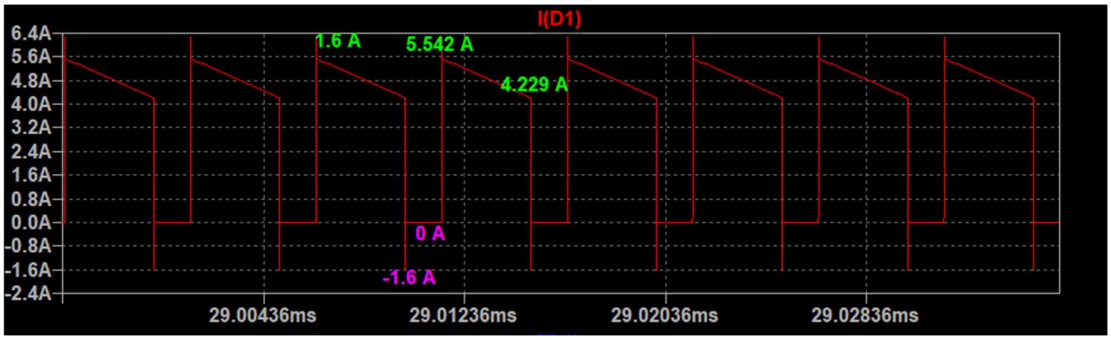
- **Switch node voltage (Vj1):** The switch node voltage toggles between approximately 42V (MOSFET ON) and -0.59V (MOSFET OFF). During off state, the node goes slightly negative to forward bias the diode allowing continuous inductor current flow. This confirms correct freewheeling operation of buck converter.



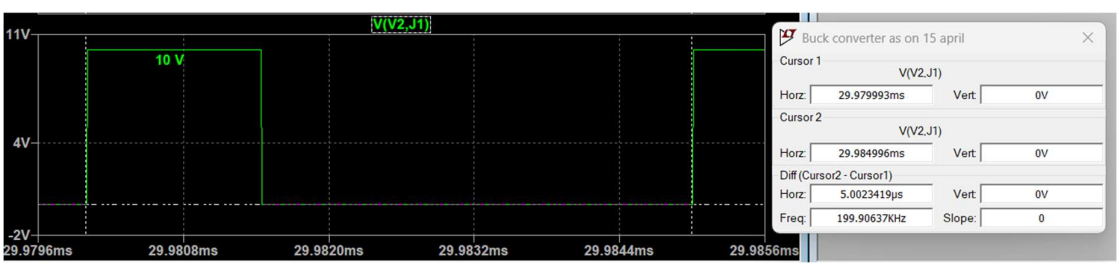
- **Diode Current (Id):** The diode current remains 0A during MOSFET on state interval. At the instant of turn-off, the diode current rises abruptly to match the inductor current as the inductor current cannot change instantaneously.

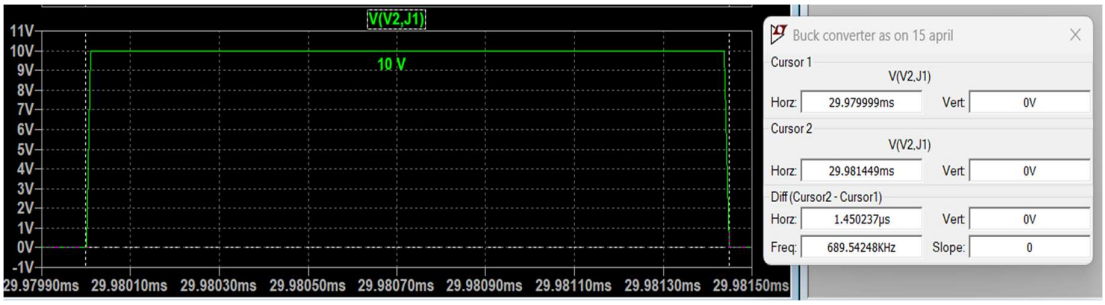
During off interval, diode current follows a downward ramp, corresponding to decreasing inductor current. Confirms correct freewheeling operation and current continuity.

- **Diode current transition behaviour:** The diode current exhibits a peak of approx. 6.4A, slightly higher than the inductor peak current due to switching transient and parasitic effects. A negative current ~-1.6A is observed during switching transitions, attributed to parasitic ringing and diode reverse recovery behaviour. These effects are typical in practical converters and highlight non-ideal switching characteristics.



- **Duty cycle:** The duty cycle of buck converter was measured from gate to source voltage waveform. The switching period(T) is measured as 5.002µs, corresponding to switching frequency 199.9063KHz. While the on time is measured as 1.4502µs. The slight deviation (~ 1–1.2%) is due to finite switching transition and cursor placement accuracy.





6. Result analysis:

- **Inductor current analysis:** The inductor current waveform exhibits a triangular ripple due to alternating voltage applied across the inductor during switching operation.
  During on state of the MOSFET, the switch node is approximately equal to the input voltage (42V), resulting in a positive voltage across inductor:
  VL = V(in)-V(out) = 42 – 12 = 30V This positive voltage causes the inductor current to increase linearly.
  While during off state, the diode conducts and switch node drops slightly below ground (~ -0.7V) resulting in VL = -(Vout+Vd) = ~ -12.7V. This negative voltage causes the inductor current to decrease linearly.
  Due to this alternating voltage the inductor current varies between 4.229A to 5.542A. Confirming continuous conduction mode (CCM) operation.
  The magnitude of the current ripple is governed by the inductor value, switching frequency and the applied volt-second balance across the inductor.

- **Switch node analysis:** The switch node voltage alternates between two states depending on switch position. When MOSFET is on, the switch node is connected to the input voltage, hence when MOSFET is off, the diode conducts and clamps the switch node close to ground potential also the inductor current must continue flowing due to stored energy, which forward biases the diode and provides a freewheeling path.
- **Output voltage ripple:** Output voltage ripple occurs due to the charging and discharging of the output capacitor by pulsating inductor current. When inductor current is greater than load current capacitor charges when it is lower capacitor discharges.
- **Duty cycle validation:** The duty cycle calculated theoretically is validated against simulation results. The small deviation is observed is due to non-ideal effects such as diode forward drop and MOSFET on state resistance.
- **Non-idealities and practical consideration:** Like MOSFET Rds(on) causing the conduction loss during on state, diode forward drop which introduces the power loss during freewheeling interval and increases off-state loss. These non-idealities result in slight deviation from ideal output voltage observed ~ 11.7V instead 12V.

7. **Calculated vs Simulated comparison:**

Parameters	Calculated	Simulated	Deviation
V out	12V	11.716 V	2.4%
IL avg	5A	4.885 A	2.3%
Δ iL	1.3A	1.31A	0.8%
Vout ripple	37mv	35mv	5.4%
Duty Cycle	28.57%	28.95%	1.3%
Frequency	200KHz	199.906KHz	0.05%
CCM Confirmation	Imin > 0A	4.229 A	CCM Confirmed

8. **Conclusion:** The buck converter design successfully steps down the input voltage with expected duty cycle control. The inductor current exhibits continuous conduction with triangular ripple and output voltage closely follows the theoretical predictions with minor deviations due to non-ideal components.